

PD Group:

Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS

BIOLOGY
Higher 2

9747/02

Paper 2 Core Paper

9 September 2008 Tuesday

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Section B

Answer any **one** question.

Write your answers on separate writing paper.

At the end of the examination,

1. fasten all your work securely together;
2. hand in your answers for Section A and Section B separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Section A

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Section B

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Section A

Answer all the questions in this section.

QUESTION 1

Fig. 1.1 is an electron micrograph of a fibroblast. Fibroblasts are flattened, irregular-shaped connective tissue cells that secrete components of extracellular matrix such as hyaluronic acid and collagen.

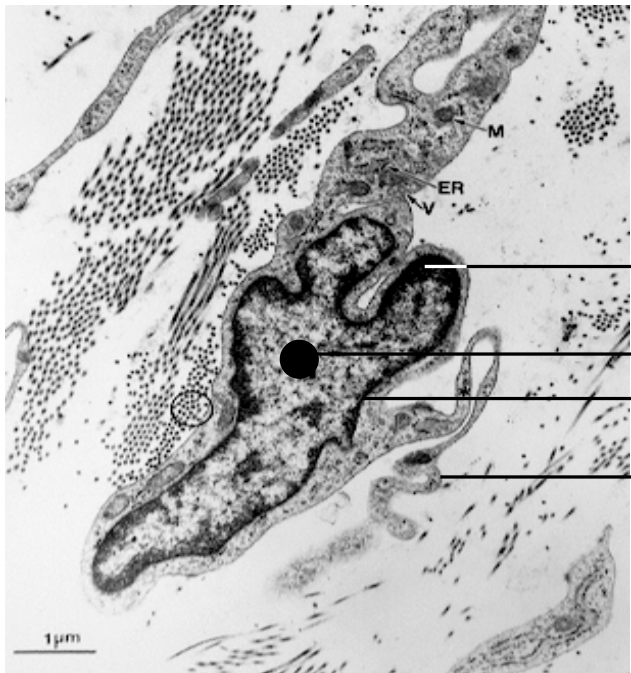


Fig. 1.1

Picture adapted from www.udel.edu/~histopage

(a) Name structures A to D.

A Heterochromatin;

B Nucleolus;

C Nuclear envelope/ membrane;

D Cell Membrane/ Plasma membrane;

[2]

(b) Explain why collagen is described as a fibrous protein.

- Insoluble in water;
- Secondary structure (helical);
- Polypeptide chains cross link to form long fibres;
- Regular amino acid sequence / every 3rd residue is a glycine;

[2]

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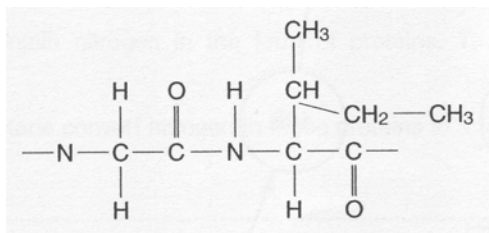
Collagen is continuously broken down in the extracellular matrix by the enzyme collagenase, which catalyses the hydrolysis of the peptide bond between the amino acids glycine and isoleucine.

(c) Suggest how collagenase is only able to act on the peptide bond between glycine and isoleucine and not on peptide bonds between any other amino acids.

- active site;
- specific complementary shape;
- only accepts/recognizes R grps of these 2 amino acids;

[2]

(d) Draw a diagram below to show how the peptide bond between glycine and isoleucine is broken down by hydrolysis including the product or products.



- Correct bond broken (between C-N);
- Involvement of water molecules in breaking the peptide bond shown clearly;
- Two amino acids with free grps as follows
-COOH/-COO⁻ and -NH₂/NH₃⁺;

[3]

(e) Suggest why collagen has great tensile strength.

- Three polypeptide chains are cross linked via H bonds
- to form triple helix of tropocollagen;

[1]

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(a) State two similarities between rough endoplasmic reticulum and golgi apparatus.

[2]

(b) State two differences between rough endoplasmic reticulum and golgi apparatus.

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(c) Explain why collagen is d

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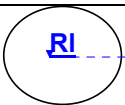
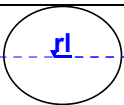
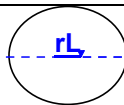
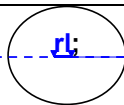
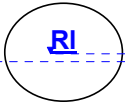
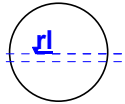
QUESTION 2

When a red flowered, three-leaved clover plant was crossed with a white flowered, five-leaved plant the following offspring were produced:

red-flowered, five leaved	36
red-flowered, three-leaved	41
white-flowered, five-leaved	40
white-flowered, three-leaved	38

The alleles for red flowers and five leaves are dominant.

- (a) Using the symbols **R** for red flowers, **r** for white flowers, **L** for five leaves and **l** for three leaves, draw a genetic diagram to explain these results. [5]

Parental phenotype	Red flowered, three-leaved clover plant,		White flowered, five-leaved plant,	
Genotype	<u>Rrll</u>		<u>rrLl</u> ;	
meiosis				
Gametes				
meiosis				
Gametes				
	<u>RrLl</u> Red flowered, five-leaved		<u>rrLl</u> White flowered, five-leaved,	
	<u>Rrll</u> Red flowered, three-leaved		<u>rrll</u> White flowered, three-leaved	
F1 genotype	<u>1 RrLl</u>	<u>1 Rrll</u>	<u>1 RrLl</u>	<u>1 rrLl</u> ;
F1 phenotype	<u>1 Red flowered, five-leaved</u> ,	<u>1 Red flowered, three-leaved</u>	<u>1 White flowered, three-leaved</u>	<u>1 White flowered, five-leaved</u> ;

- (b) When these plants were self-pollinated, explain why only the white-flowered, three-leaved plants bred true. [2]

- white-flowered, three-leaved plants were double homozygous recessive;
- All the other plants would have been heterozygotes;

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The inheritance of coat colour in a type of leopard was determined to be due to two gene loci each with alleles which have a dominant-recessive relationship. Leopards with black coats, white coats or spotted coats have been observed.

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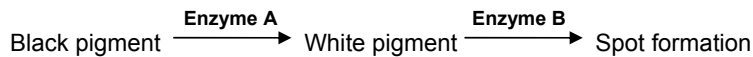
When a pure-breeding white leopard was crossed with pure-breeding black leopards, the F₁ progeny obtained were all spotted. When the spotted progeny were mated with each other, the results of the F₂ generation were

93 spotted

44 black

28 white

The suggested pathway for the synthesis of pigments involved in the development of coat colour is shown below.



(c) Explain the results of the F₂ generation.

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- Phenotype ratio should be 9: 3: 3: 1, but ratio obtained was 9 spotted: 3 white: (3+1) black;
- A B genotypes give spotted as dominant alleles give functional enzymes A & B = spotted;
- aa genotypes remain black because A enzyme is not produced;
- A bb genotypes are white because B enzyme not produced;

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QUESTION 3

One way of investigating relationships between different primate groups is to study differences in the amino acid sequences of proteins such as haemoglobin.

(a) Explain the origin of differences in the amino acid sequences in a protein.

- caused by point mutation;
- changes the sequence of DNA bases / changes nucleotide / codon;
- new triplet codes for a different amino acid;
- insertion/deletion causes frameshift;
- codes for new polypeptide;

[2]

The β chain of haemoglobin has been investigated and shown to have only four points of difference in all primate groups. The amino acids found at these positions in certain primates or primate groups are shown in the table below.

Primate or primate group	Amino acid at position 80	Amino acid at position 87	Amino acid at position 109	Amino acid at position 125
Chimpanzees	Aspartic acid	Threonine	Arginine	Proline
Gibbons	Asparagines	Lysine	Arginine	Glutamine
Old World monkeys	Aspartic acid	Glutamine	Lysine	Glutamine
New World monkeys	Aspartic acid	Glutamine	Arginine	Glutamine
Lemurs	Aspartic acid	Glutamine	Threonine	Alanine

Table 3.1

(b) (i) With reference to Table 3.1, state the two groups that seem to be closely related and give reasons for your answer.

- Old World monkeys and New World monkeys;
- Only one difference / three common amino acids;

[2]

(ii) Gibbons are usually classified with the chimpanzees as apes, in the Hominoidea. Explain why the results shown do not support this classification.

- Gibbon more like New World monkey;
- Gibbon β chain has only one common amino acid (arginine) with chimpanzee;
- 2 common amino acids with New World monkeys;
- Chimpanzee has only common amino acid with / 3 different from each of the other groups;

[3]

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(c) Describe four ways, other than differences in amino acid sequences, in which evolutionary relationships between hominids can be investigated. [4]

- ref to homology / skeletal / skull / dentition details;
- dating of fossils allows us to determine the age of a formerly living thing;
- anatomical comparisons of fossil primates;
- anatomical comparisons of the skeletons of modern species;
- sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
- antibody-antigen / immunological / serum studies;
- use of antibodies / agglutination/ immuno-precipitation;
- comparison of early stages if animal development reveals additional anatomical homologies not visible in adult organisms;

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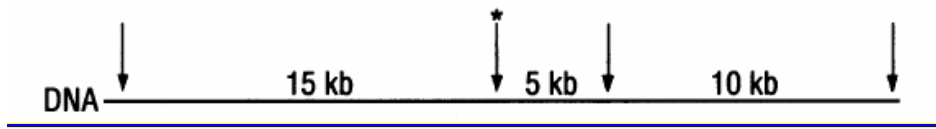
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QUESTION 4

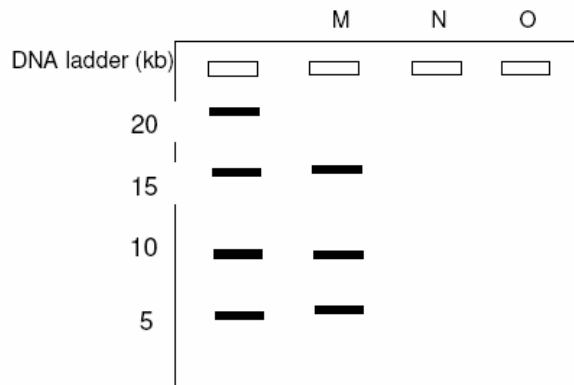
Medical scientists can now diagnose hundreds of human genetic disorders before the onset of symptoms. For example, DNA from fetuses can be isolated and screened for the presence of a diseased allele, even before the birth of the child.

A length of DNA from one of a pair of homologous chromosomes is shown in Fig 5.1. The target sites of EcoR1 are shown by arrows and the length of DNA between the target sites is given in kilobases (kb)

**Fig. 4.1**

A mutation alters one base of the restriction site marked by an asterisk (*) and individuals with that particular mutation in both alleles suffer from a childhood disease.

DNA from three fetuses were isolated and cut with EcoR1 and the DNA fragments separated by electrophoresis. Lane M shows the DNA from the fetus which is homozygous for the normal allele.

**Fig. 4.2**

(a) Describe briefly how DNA fragments may be separated by gel electrophoresis. [5]

- After digesting the DNA sample with restriction enzymes;
- the mixture of DNA is loaded in a well at one end of an agarose gel;
- submerged in aqueous buffer solution in the gel electrophoresis chamber;
- Electrodes are attached to both ends of the chamber and an electric current is applied;
- The DNA molecules which are negatively charged (phosphate groups present on DNA gives a net negative charge) migrates towards the positive electrode, the anode;
- The rate of movement traveled in a given time depends on the size of nucleic acid fragment OR Larger molecules move slower (vice versa);

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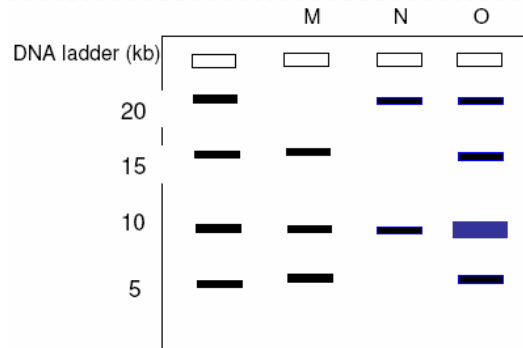
DNA

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(b) Draw (in Fig. 4.2) the bands that would be obtained for

(i) Lane N: fetus which is homozygous for the diseased allele

(ii) Lane O: fetus which is heterozygous for the diseased allele



(c) State how doctors analyze the DNA fragments to identify the fetus with the genetic defect.

- The DNA fragment from the fetus with the genetic defect has 2 bands, at 10 and 20kb;

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(ii) Lane O:

(c) State how doctors analyze the DNA fragments to identify the fetus with the genetic defect. [1]

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Section B: Free Response Questions
Answer only **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

5. (a) Outline the role of NAD and NADP in cells.

NADP/ NAD as

- Electron/hydrogen carrier or coenzymes;
- Carry hydrogen atoms from one compound to another;

NADP in photosynthesis

- Final electron acceptor in non-cyclic photophosphorylation;
- Drives photolysis of water;
- $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
- NADPH passes to Calvin cycle/ light independent reactions in the stroma;
- NADPH reduces glycerate biphosphate/1,3 diphosphoglycerate to triose phosphate/ glyceraldehyde phosphate and gets reoxidized back to NADP;

NAD in respiration

Aerobic respiration:

- Glycolysis: Accept H^+ from triose phosphate/ glyceraldehyde phosphate, which is oxidized to 1,3 diphosphoglycerate or pyruvate;
- Forms 2 NADH for each glucose molecule;
- Link reaction: Accept H^+ from pyruvate, which is converted to acetyl CoA;
- Forms 2 NADH for each glucose molecule;
- Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;
- Forms 6 NADH for each glucose molecule;
- Link rxn and Krebs cycle occurs in the mitochondrial matrix;
- Carry hydrogen/ electrons to ETC;
- Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
- 1 NADH yields 3 ATP;
- NAD reoxidised;

Anaerobic respiration:

- NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;
- NAD reused in glycolysis to generate 2 ATP;

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(b) Describe how photophosphorylation differs from oxidative phosphorylation.

[8]

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Feature	photophosphorylation	oxidative phosphorylation
Organelle	Chloroplast	Mitochondria
Location	Thylakoid membrane	Inner Mitochondrial membrane
Proton reservoirs	Thylakoid space	Intermembrane space
Energy conversion	Light to chemical	Chemical to chemical
Role of coenzymes	NADP is reduced in the light rxn as final electron acceptor	NAD is reduced in glycolysis, link rxn and Kreb's cycle
Sources of electrons	Water	NADH and FADH
Roles of O ₂	By-product of photolysis	Final electron acceptor
Roles of water	Photolysis: To produce H ⁺ which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre	By-product of O ₂ as the final electron acceptor

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HCI Prelim 2007¶

6. (a) Describe the structure and function of the portions of eukaryotic DNA that do not encode for protein or RNA. [12]

- In eukaryotes, about 97% of DNA does not code for proteins or RNA;
- They consist of regulatory sequences, introns and repetitive DNA;

UTR

- UTRs are untranslated regions found at the 5' and 3' ends of most mRNAs;
- 5' UTR is also known as a leader sequence;
- contains a ribosome binding site;
- contains binding sites for proteins that affects mRNA's stability;
- prokaryotic mRNA's 5' UTR is usually short;
- some viruses have long 5' UTR which may be important in gene expression;

Promoter

- DNA sequences found at the 5' end adjacent to transcriptional site;
- immediately upstream of coding region of gene;
- RNA polymerase and transcription factors bind to the promoter;
- To initiate production of mRNA transcript;
- Interactions of proteins at the promoter regulate gene activity;
- By activating or repressing transcription;
- An eg. of a 6 promoter seq. is the "TATA box"/ TATAAA;
- Found about 25-35 nucleotides before the transcriptional site;

Control elements

- Are noncoding DNA segments that regulate transcription by binding transcription factors;
- Proximal control elements are located close to the promoter;
- Distal control elements or enhancers may be far away from a gene;
- enhancer / silencer sequences have base sequences that can be recognized by activator / repressor proteins respectively;
- interaction with promoter occurs via protein-mediated DNA bending;

Repetitive DNA

- about 10% of mammals' genome is tandemly repetitive DNA/ satellite DNA;
- much of the satellite DNA play a structural role at telomeres and centromeres;
- Telomeres - Protects the ends of chromosomes from hydrolysis by cellular exonucleases;
- Centromeres - region where sister chromatids join in a replicated chromosomal structure during cell division;
- anchor point for spindle formation required for the pull of chromosomes toward the centrioles during anaphase;
- allows for equal separation of genetic material to daughter cells during anaphase of cell division;
- absence of centromere results in nondisjunction of chromosomes; aneuploidy;

Transposons

- Genes that move from one location to another within the genome;
- If it "jumps" into a coding sequence, it can prevent normal gene function;

[1 mark each]

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Deleted: Explain how homology (anatomical, embryological, molecular) supports Darwin's theory of natural selection.

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<#>ref. examples: ¶
forelimbs of all mammals, including humans, cats, whales and bats show the same arrangement of bones from the shoulder to the tips of the digits/ vestigial legs in snakes¶
<#>even though these appendages can have very different functions (lifting, walking, swimming ... [228]

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(b) Explain how the substitution of one nucleotide in DNA can be selected for or against in an area endemic with malaria.

- A point mutation results in the incorporation of hydrophobic valine instead of hydrophilic glutamic acid in haemoglobin;
- Normal haemoglobin (haemoglobin A) in red blood cells is replaced entirely by abnormal haemoglobin (haemoglobin S);
- Individuals who are homozygous $H^A H^A$ = Normal, homozygous $H^S H^S$ = sickle cell anaemia, heterozygous $H^A H^S$ = sickle cell trait;
- 2 strong selective pressures acting on H^A and H^S ;
- Selection against people homozygous for the sickle cell allele ($H^S H^S$) is strong since they become seriously anemic;
- Selection against people homozygous for the normal allele ($H^A H^A$) is strong as they are likely to die from malaria;
- However, heterozygous people are at a selective advantage;
- they do not suffer from sickle cell anemia and are less likely to suffer badly from malaria;
- The parasite *Plasmodium* is less well adapted to living in r.b.c with haemoglobin S;
- Parasites that invade r.b.c with haemoglobin S will die;
- trapping the parasites within them;
- Infection is stopped;

[Any 8 – 1 mark each]

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QUESTION 9

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3	Presence of i - presence of i - function as y gene regulati
4	Monocistroni - individual en by separate p - can be produ at all; - hence allowi intermediates to suit the cell:
5	Presence of c - fine tuned co for different "le gene dependin present in the of the gene's - response to (Accept: trans

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Cell structure + Biomolecules + Enzymes
June 2005 Paper 2 Q4

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VJC Promo 2006 Q1d

(a) State two similarities between rough endoplasmid reticulum and golgi apparatus.

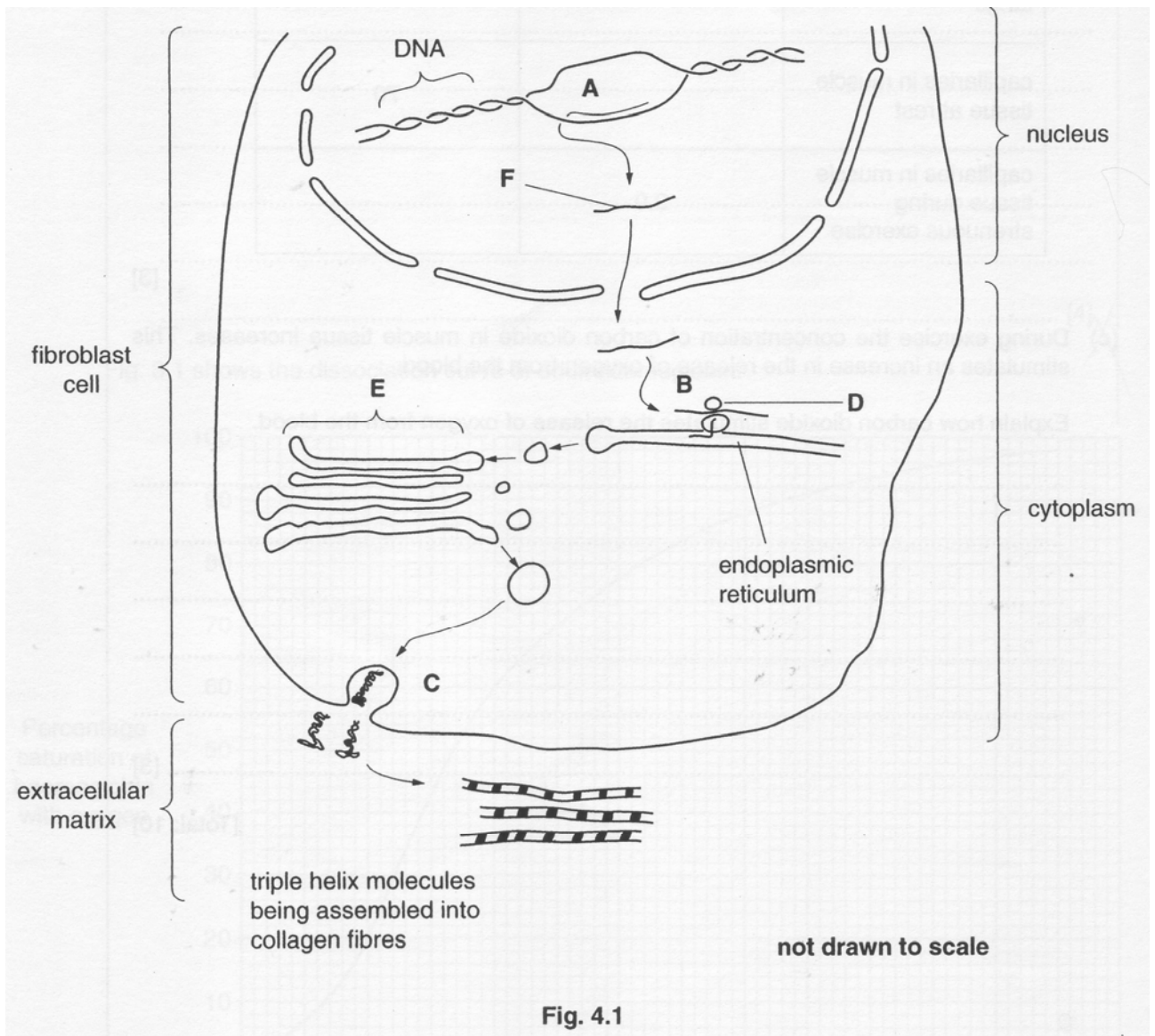
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(b) State two differences between rough endoplasmid reticulum and golgi apparatus.

[2]

Similarity (<i>any one</i>) - Both consist of membrane-bound sacs/tubules called cisternae. [1] - Both are made up of single membrane; [1] - Both produce vesicles that bud off the organelle; [1/2]	
Differences	
Structure D (rER)	Structure F (GA)
1. Consist of a <u>network</u> of <u>interconnected</u> membrane-bound sacs and tubules. [1/2]	Consist of <u>stacks</u> of <u>flattened</u> membrane-bound sacs. [1/2]
NB: idea of interconnectedness is impt.	
2. Presence of ribosomes. [1/2]	Absence of ribosomes. [1/2]
® ribosomes <i>may or may not</i> be present	
3. Connected to the outer membrane of the nucleus. [1/2]	3. Not connected to the outer membrane of the nucleus. [1/2]
4. Does not have a definite forming and maturing face.	4. Has a definite forming face where the vesicles join and a maturing face where the vesicles bud off.

Fig. 4.1 is a diagram summarizing these events.



(a) (i) Name the processes occurring at **A**, **B** and **C**. [3]

A Transcription
B Translation
C exocytosis

(ii) Name structures **D** and **E**. [2]

D Sub unit of ribosome
E Golgi apparatus/body

(iii) Name molecule **F**. [1]

F mRNA

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a- to remove diagram and combine with something else....

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Nov 2003 Paper 2 Q1

- (c) Explain why collagen is described as a fibrous protein.
[2]

insoluble nature of collagen;
secondary structure – three polypeptides chains are cross-linked via
hydrogen bonding to each other to form a triple helix;
Fibrils like parallel in bundles to form a collagen fiber;

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VJC Promo 2006 Cell and Nuclear Division

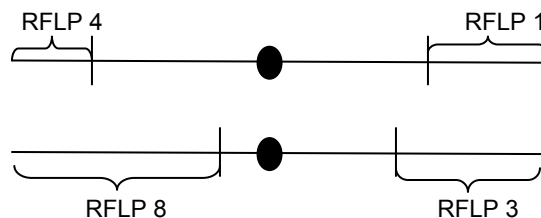
Cell and Nuclear Division

RJC Prelim 2007

QUESTION 2

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a male.



An offspring of the male has a chromosome with RFLPs 1 and 8, contributed by the father.

- (a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8.
[1]

-
- (ii) In the space below, draw the event occurring in (a).
[3]

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations. [1]

The micrograph below shows a Lily cell undergoing cell division.



[© Clare Hasenkampf/BPS]

(b) Refer to the micrograph above and:

(i) Identify the type and stage of nuclear division.

[2]

(ii) Briefly explain the reason for your answer in (b)(i).

[2]

(c) Describe the roles of spindle fibers in meiosis.

[2]

Tubulin is involved in the construction of microtubules.

(a) State where microtubules originate in the cell.

_____ [1]
]

(b) Describe the role of microtubules in mitosis.

_____ [1]
]

The chromosome drawings show four stages in mitosis.



(c) Give the correct sequence of the drawings.

_____ [1]
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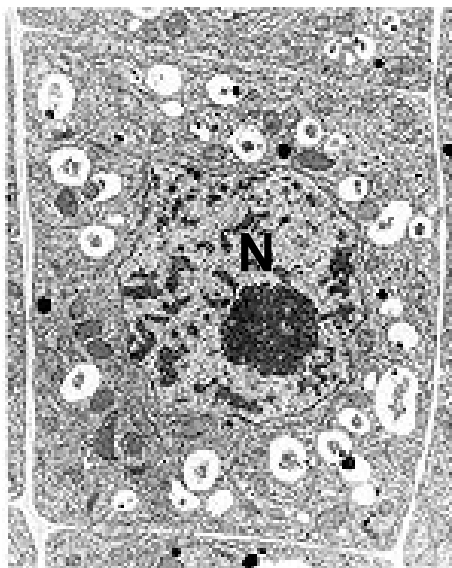
(d) What is the diploid number of the chromosome in these cells.

_____ [1]
]

(e) If you look at Figure Z, the chromosomes appear to be double structures. Suggest why?

_____ [1]

The DNA in the nucleus (**N**) of the plant cell in the electron micrograph, Figure 5.1, below appears different from that of a cell undergoing mitosis (Figure 5.2).



(http://www.museum.vic.gov.au/scidiscovery/images/mn003613_w150.jpg)
(http://plantphys.info/Plant_Biology/images/anaphase1.jpg)

Figure 5.1



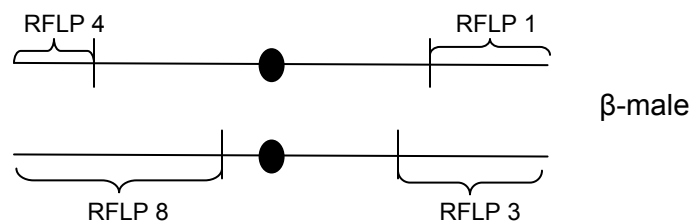
Figure 5.2

Account for this difference.

[Total: 11]

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a β -male.

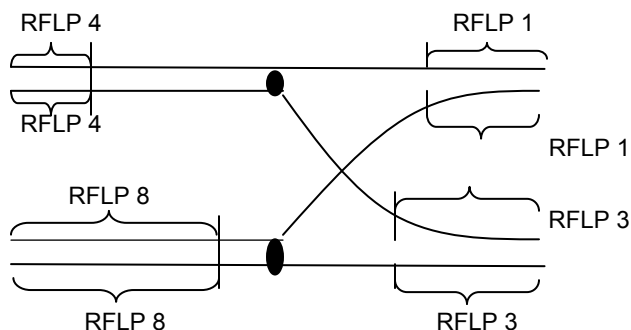


An offspring of β -male has a chromosome with RFLPs 1 and 8, contributed by the father.

(a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8. [1]

crossing over

(ii) In the space below, draw the event occurring in (a). [3]



[1] - 2 sister chromatids per chromosome
[1] - a crossing over of non-sister-chromatids; may also take place on the other arm between RFLP 4 and RFLP 8
labelling of centromere (1/2) and chromatids/chromosomes (1/2)

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations [1]

Because of its late onset

Page Break

The micrograph below shows a lily cell undergoing cell division.



[© Clare Hasenkampf/BPS.]

- (b) In reference to the micrograph above, identify:

- (i) The type and stage of nuclear division [2]

Meiosis

- (ii) Briefly explain the reason for your answer in (b) (i). [2]

Metaphase bec:

metaphase bec chromosomes are aligned at equator

metaphase 2 bec:

2 metaphase plate has formed → meiosis

OR

the daughter cell have chromosomes aligned at the new equatorial plate which is at right angles to the first

- (c) Describe the roles of spindle fibres in meiosis. [2]

pull sister chromatids/homologous chromosomes to opposite poles at anaphase
help align chromosomes at equatorial plate

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Other (non-kinetochore) spindle fibres lengthen to elongate the cell in preparation for cytokinesis

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- (a) Centrosome / microtubule organizing centre (MTOC) / centrioles; [1]
- (b) Form spindle fibres / spindle;
For separating sister chromatids; (**DO NOT** mention "**splitting** of sister chromatids")
Equal distribution of chromosomes;
(1/2 each; 1 max)
- (c) Y, X, Z, W [1]
- (d) 6 [1]
- (e) During S phase of interphase, replication of DNA occurs;
Condensation of DNA molecules tightly around histones;
Two DNA molecules appear as sister chromatids in prophase;
(1/2 each; 1 max)
- (f) Account for this difference. [3] (1/2 each; 3 max)

Figure 5.1:
DNA as chromatin;
Long, thin / not visible;
DNA wound around histones;

Figure 5.2:
DNA replicated during interphase;
Highly condensed DNA by spiralisation;
Chromosome consists of two sister chromatids;
Sister chromatids separate and move, centromeres first, to opposite poles of spindle fibres;

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RJC promo 2006 DNA & Genomics

Genetics of bacteria

TJC Prelim 2007

QUESTION 3

Bacteria can regulate the expression of its genes in response to changes in the environment, typically to the presence or absence of nutrients. Some bacterial genes

encode for regulatory proteins which controls the expression of other genes or operons.

For example, *Escherichia coli* can utilize any sugar as its source of carbon and energy.

However, when grown in a medium containing both glucose and lactose, the bacteria will

use up one before the other. This preferential use of carbon source is said to be governed

by the “**dual control of the *lac* operon**”.

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what is meant by “dual control of the *lac* operon”

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An experiment was conducted to examine the effects of glucose and lactose on the levels

of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals.

The results are shown in Fig. 1 below.

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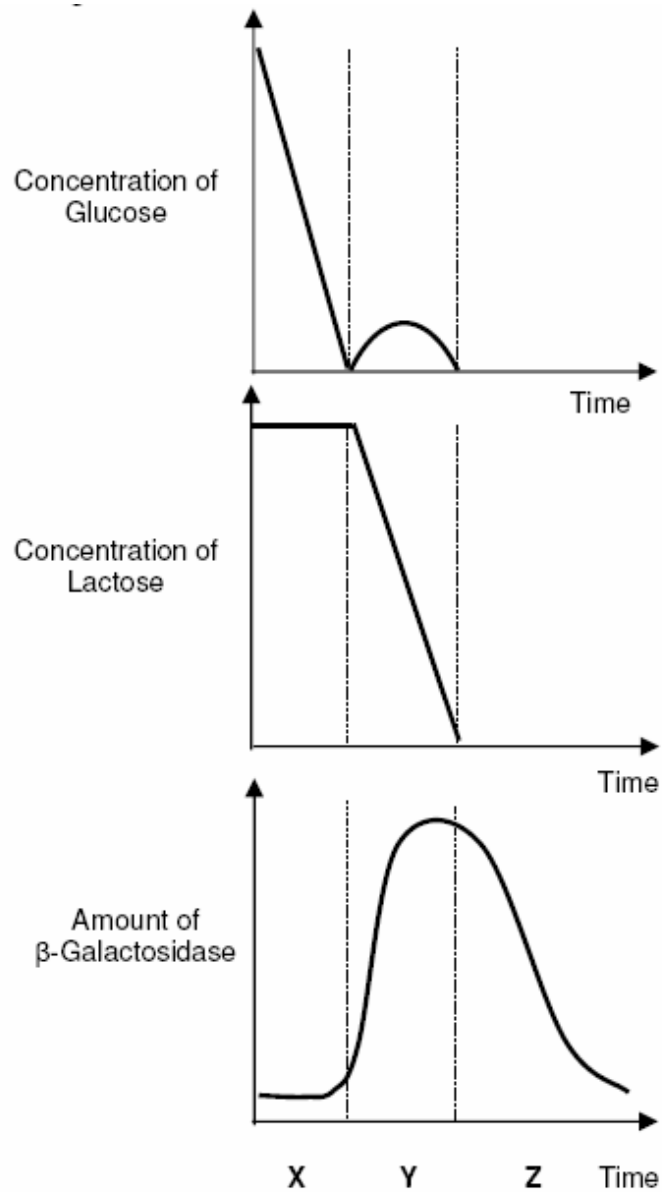


Fig.1

X:

Both glucose and lactose are present and only glucose is used;
 In the presence of lactose, allolactose binds to lac repressor;
 Thus, the Lac repressor is in its inactive configuration and does not
 bind to the operator;

But another mechanism prevents transcription of the lac operon due to catabolite repression (ie. CAP is inactive and disengages from CAP binding site);

RNA polymerase is unable to bind to the promoter/Low β – galactosidase mRNA is produced;

[Any 3 - 1 mark each]

Y:

Once the glucose concentration has depleted, *E. coli* uses Lactose; Allolactose binds to the lac repressor;

Lac repressor is inactive and does not bind to the operator;

RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lac operon / Lac operon is switched on / High β – galactosidase mRNA is produced;

[Any 3 - 1 mark each]

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suggest one mechanism that causes the amount of β -galactosidase to decrease.

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State the two mechanisms of regulation of lac operon

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negative control by the lac repressor;
positive control by cAMP receptor protein;

[2]

[1 mark each]

(b) With reference to Fig.1, explain how expression levels of β

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-galactosidase is regulated by the lac repressor from time X to Y.		
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X:		
Y:		
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(c)

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Describe another mechanism which regulates the expression of β -galactosidase at Y.		
[2]		

(d)

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mRNA degradation/ protein is degraded.		
[1]		
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- (ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source. [1]

- (a) State the two mechanisms of regulation of lac operon. [2]
negative control by the lac repressor
positive control by cAMP receptor protein.

(b) With reference to Fig.1, explain how expression levels of β -galactosidase is regulated by the **lac repressor** from time X to Y. [6]

X:

Both glucose and lactose are present and glucose is used but not lactose
In the presence of lactose, allolactose binds to lac repressor. Thus, the Lac repressor
is in its inactive configuration and does not bind to the operator
But another mechanism prevents transcription of the lac operon due to catabolite
repression (ie. CRP is inactive and disengages from CRP binding site)
RNA polymerase is unable to bind to the promoter/Low β -galactosidase mRNA is produced

Y:

Once the glucose concentration has depleted, *E. coli* uses Lactose.
Allolactose binds to the lac repressor
Lac repressor is inactive and does not bind to the operator
RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lac
operon / Lac operon is switched on / High β -galactosidase mRNA is produced

(c) Describe another mechanism which regulates the expression of β -galactosidase at Y. [2]

In low levels of glucose, cAMP levels increases , CRP is active and it binds to the CRP binding site.
Transcription of B-gal proceeds at a high level.

(d)(i) Suggest one mechanism that causes the amount of β -galactosidase to decrease. [1]

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mRNA degradation/ protein is degraded

(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source. [1]

Stress genes can be turned on which allows the bacterial cells to survive.

Under

stress or starvation (starvation sigma factor)

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In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly (A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP).

The table below illustrates the effects of capping and polyadenylation on the lifespan of the luciferase mRNA and its protein products.

Table 3.1

mRNA	Luciferase mRNA Half Life (min)	Luciferase Activity (units)
Uncapped		
Poly(A) tail absent	31	2,941
Poly(A) tail present	44	4,480
Capped		
Poly(A) tail absent	53	62,595
Poly(A) tail present	100	1,331,917

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- a) During a space probe to Mars, the genetic system of a newly discovered bacterium was brought back to your laboratory. After running several tests on it, the following information was obtained:

The genetic material of the bacterium is composed of nucleotides, each having one of the six nitrogenous bases: guanine (G), adenine (A), thymine (T), cytosine (C), and two other unknown bases, which you name W and X. The bacterium was also found to have base composition: 24% G, 19% A, 19% T, 24% C, 7% W and 7% X.

- i. Based on the above information, what can be deduced about the likely structure of the genetic material of the bacterium? [2]

Genetic material consists of a *double stranded* polynucleotide chain [1]

G pairs with C, A pairs with T and W pairs with X [1] Pairing – all or none

The bacterium has 50 essential amino acids and 16 non-essential amino acids.

- ii. What is the minimum number of nucleotides that code for one amino acid in this bacterium? [1]

3 [1] (6x6x6= 1295)

- b) Haemoglobin C (Hb C) is a variant of adult haemoglobin (Hb A) that has arisen by mutation in the β globin sequence and patients with recessive Hb C disorder exhibit mild hemolytic anemia.

Table 1. The first 8 amino acids of the β globin sequence of Hb A and Hb C

Hb A β globin	val – his – leu – thr – pro – glu – glu –lys.....
Hb C β globin	val – his – leu – thr – pro – lys – glu –lys.....

Table 2 The genetic code (the three bases of an mRNA codon are designated here as the first, second and third bases, reading in the 5' $\rightarrow$ 3' direction along the mRNA)

		Second base				
		U	C	A	G	
First base	U	UUU phe	UCU ser	UAU tyr	UGU cys	U C A G
		UUC phe	UCC ser	UAC tyr	UGC cys	
		UUA leu	UCA ser	UAA stop	UGA stop	
		UUG leu	UCG ser	UAG stop	UGG trp	
	C	CUU leu	CCU pro	CAU his	CGU arg	U C A G
		CUC leu	CCC pro	CAC his	CGC arg	
		CUA leu	CCA pro	CAA gln	CGA arg	
		CUG leu	CCG pro	CAG gln	CGG arg	
	A	AUU ile	ACU thr	AAU asn	AGU ser	U C A G
		AUC ile	ACC thr	AAC asn	AGC ser	
		AUA ile	ACA thr	AAA lys	AGA arg	
		AUG met	ACG thr	AAG lys	AGG arg	
	G	GUU val	GCU ala	GAU asp	GGU gly	U C A G
		GUC val	GCC ala	GAC asp	GGC gly	
		GUA val	GCA ala	GAA glu	GGA gly	
		GUG val	GCG ala	GAG glu	GGG gly	

Use the information in **Tables 1 and 2** to answer the following questions.

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E = expected 'value'

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Phenotype ration should be 9: 3: 3: 1, but ratio obtained was 9 spotted: 3 white: (3+1) black [1]

A_B _ genotypes give spotted as dominant alleles give functional enzymes A& B = spotted [1]

aa _ _ genotypes remain black because A enzyme is not produced [1]

A_ bb genotypes are white because B enzyme not produced [1]

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Answers

- (a) caused by point mutation;
changes the sequence of DNA bases / changes nucleotide / codon;
new triplet codes for a different amino acid;
substitution causes frameshift;
codes for new polypeptide [2]

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- (b)(i) Old World monkeys and New World monkeys;
Only one difference / three common amino acids [2]

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- (b)(ii) Gibbon more like New World monkey;
Gibbon β chain has only one common arginine / amino acid with chimpanzee;
2 common amino acids with New World monkeys;
chimpanzee has only common amino acid with / 3 different from each of the other groups [3]

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- (c) anatomical comparisons of fossil primates;
dating of fossils;
sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
anatomical comparisons of the skeletons of modern species;
ref to homology / skeletal / skull / dentition details;
antibody-antigen / immunological / serum studies;
use of antibodies / agglutination/ immuno-precipitation; [any 2]

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modify the peak for fig. 6.2 at 1st part...

- 6** What do you understand by the following terms?
- (a) i. *Species* [2]
- Any group of organisms;
 - Capable or potentially capable of interbreeding with one another;
 - To produce fertile offspring;
 - Show any similarities in evolution, ecology, morphology, behavior, biochemistry (any 2);
- ii. *Genetic variation* [1]
- Members of a population;
 - Differ in one or more inherited characteristics;
- (b) What impact would low genetic variability in a population of this species have on species survival? Explain your answer. [2]
- Low number of alleles would reduce probability of the presence of a favorable characteristic in the population;
 - If there was an environmental change;
 - Reduced chances of survival / species extinction;
 - Continuous inbreeding between closely related individuals;
 - Inbreeding depression may occur;
- (c) Using the information from **Figure 6.1**, explain the variation shown. [2]
- Polygenes;
 - Continuous variation / normal distribution / bell shaped;
 - Range seen due to the additive effects of genes / alleles;
 - Effects of environment;
- (d) How does the information in **Figure 6.2** support your answer in (c)? [1]
- Breeding with individuals of similar genotypes;
 - Shifted the mean / variation towards either end;

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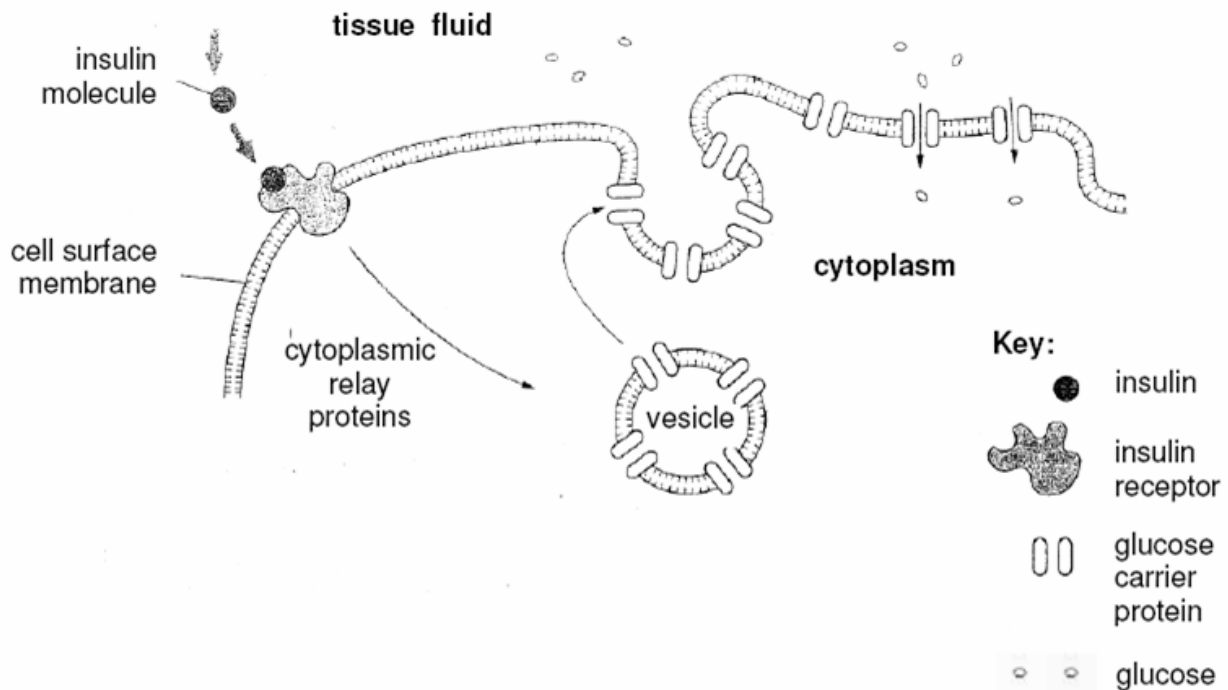
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Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.

Fig.

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6.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.



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Fig. 6.1**(a)**

State the source of the insulin

molec
ule in
Fig.

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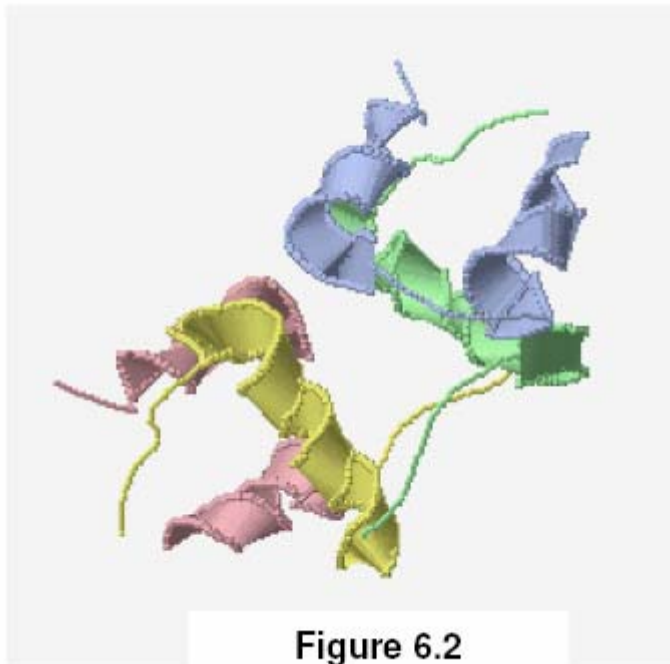
6.1. [1]

(b)
The
insulin
recept
or is a
protei
n.

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be a genetic risk factor for Type 2 diabetes.
[1]

The structure of the insulin receptor is shown in Figure 6.2.



(c) With reference to Figures 6.1 and 6.2,

(i)
explain how the binding of insulin to its receptor results in activation of tyrosine kinase

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explain how the binding of insulin to its receptor results in an increase in the

phosphorylation of serine and threonine residues side-chains of the cytoplasmic relay proteins.

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[3]

(ii)

describe the events

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of serine and

threonine residues of cytoplasmic relay proteins

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that lead to an increase in the number of glucos

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rane.

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[2]

(d) Type 2 diabetes patients are not able to regulate their blood glucose levels owing to the loss of their cells' sensitivity to insulin.

(d) (i) If an increase in blood glucose concentration is not corrected for some time, symptoms of diabetes such as excessive thirst may occur. Suggest how excess blood glucose may cause excessive thirst.

[1]

(ii) Explain how the blood insulin level is homeostatically regulated in a normal

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individ
ual.

[3]

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Answer:

(a)

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Identify the source of the insulin molecule in **Figure 6.1**. [1]

Beta

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cells of the pancreatic islets of Langerhans;;
(**Reject:** b-pancreatic cells / b-islets)

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(b) The insulin receptor is a protein.

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be a genetic risk factor for Type 2 diabetes.

[1]

Harmful mutation of the **gene encoding** the **insulin receptor** (reject: insulin) may occur;

(Accept: reference to harmful effects of the mutation eg. alteration of specific 3-D conformation of insulin receptor or defective / non-functional / malfunctional receptor;

Reject: genetic defect / defective gene / faulty gene ie word "mutation" not mentioned.) mutant allele / gene is **transmitted to / inherited by future generations**;

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With reference to **Figures 6.1** and **6.2**,

(i)

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explain how the binding of insulin to its receptor results in activation of tyrosine kinase

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in an increase in the phosphorylation of serine and threonine residue side-chains of cytoplasmic relay proteins

[
3
]

1]
X:
Y:

(b) Explain the significance of Y in electrical transmission of nervous impulses. [3]

Often present at the synaptic knob of many motor neurons is a large number of mitochondria.

(c)(i) State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor neurone. [2]

(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases. [2]

(d) Predict the effect on an action potential when the external concentration of sodium ions are lowered. [2]

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Nervous Control
TJC Prelim 2007

QUESTION 7

Genetic modification of crop plants can be carried out with the usage of a plant cloning vector known as the Ti plasmid. This plasmid is found in cells of the bacterium *Agrobacterium tumefaciens*, which normally lives in soil. When these bacteria infect a plant cell, a 30,000-base pair segment of the Ti plasmid, known as the T DNA sequence, separates from the plasmid and incorporates into the host cell genome. Insertion of a foreign DNA sequence into the T DNA sequence of the Ti plasmid hence allows for the insertion of that foreign DNA sequence into the plant genome. The process is shown in **Figure 4.1** below.

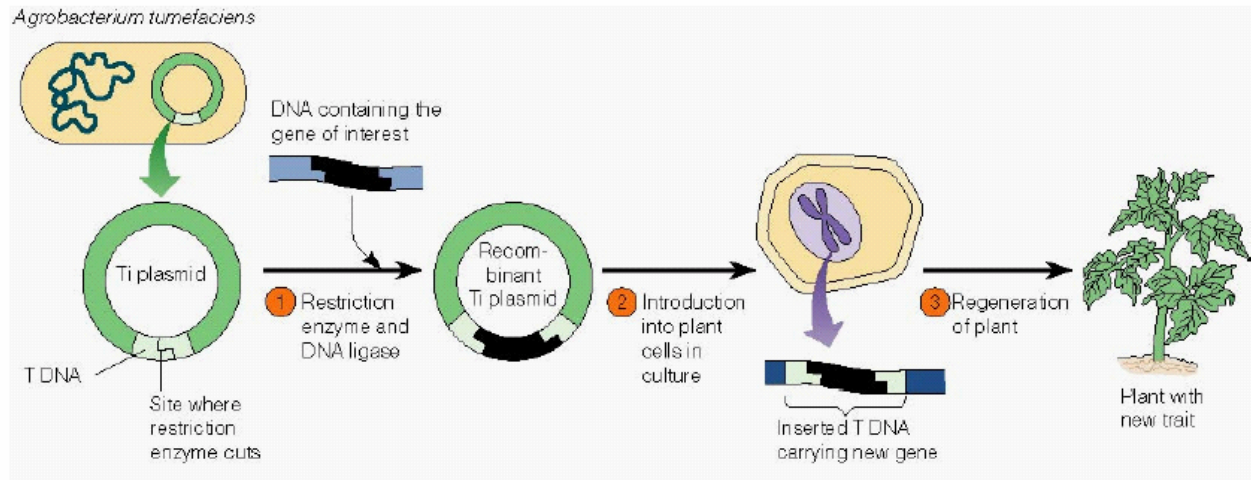


Figure 4.1

(a) Explain why it is important that the gene of interest be inserted into the T DNA sequence of the plasmid. [2]

(b) The usage of complementary DNA, rather than genomic DNA, is generally preferred in a situation where the host cell is a bacterial cell rather than a plant cell. Suggest why this is so. [2]

Extensive field trials have been carried out to assess the environmental effects of growing genetically modified (GM) crop plants, such as maize and oilseed rape. The crop plants have been genetically modified so that they are resistant to certain herbicides.

(c) Why is it important to assess the environmental effects of growing GM crops? [2]

Each crop was planted in a number of different sites. One half of each site was planted with a non-GM crop and the other half was planted with the GM crop. The crops were grown for three years and during this time many different measurements were made. Measurements included the density of weeds in the crops and the numbers and types of invertebrates, such as predatory insects, living on and in the soil. **Table 4.1** shows some of the summarized results of these field trials.

Factor	GM maize compared with non-GM maize	GM oilseed rape compared with non-GM oilseed rape
Weed density	Increase	Decrease
Number of weed seeds in soil	No change	Decrease
Number of invertebrates on soil surface	Increase	Decrease

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Figure 2 is an electron micrograph which shows the axon of a motor neuron in transverse section.

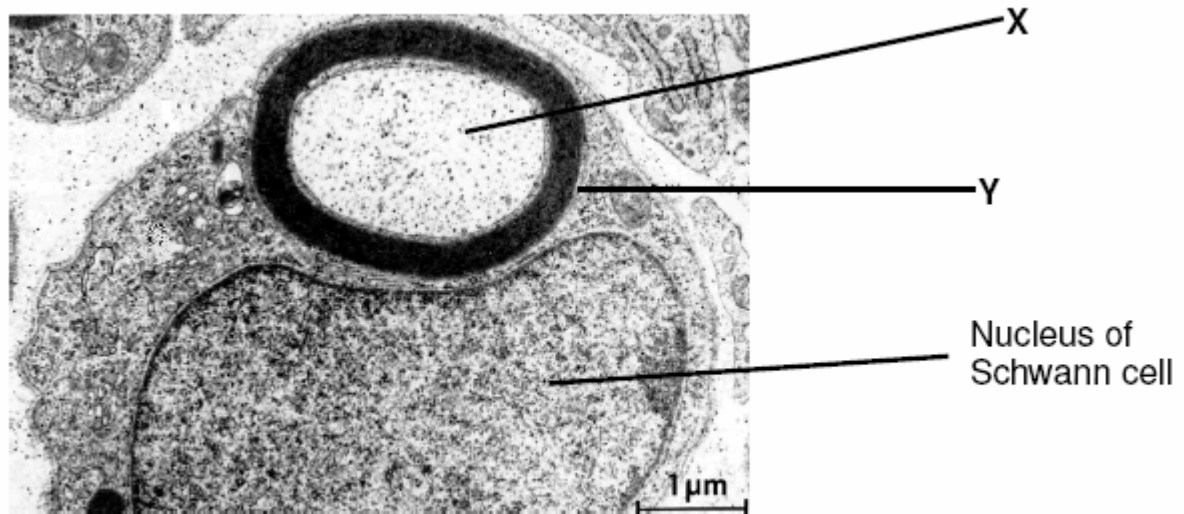


Fig. 2

(a) Label the structures X and Y.

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(d) With reference to **Table 4.1**,

- (i) state which GM crop seemed to be less harmful to the environment. [1]
- (ii) give a reason for your answer in (b)(i). [2]
- (iii) discuss which is a more commercially viable crop for the farmer to grow. [2]

The scientists further investigated the transfer of the introduced gene from the GM maize and oilseed rape plants to other plants.

(e) Suggest how the transfer could have taken place. [1]

[Total: 12]

(a) Explain why it is important that the gene of interest be inserted into the T DNA sequence of the plasmid. [2]

T DNA is inserted into the plant genome; Hence allowing for **foreign gene** within it to **also be inserted into the plant genome;** **Stable** expression (vs transient expression); **Passed on to all daughter cells** of the transformed cell;

(b) The usage of complementary DNA, rather than genomic DNA, is generally preferred in a situation where the host cell is a bacterial cell rather than a plant cell. Suggest why this is so. [2]

complementary DNA does not contain introns; use of genomic DNA necessitates the **removal of introns;** **which could not be carried out by a bacterial cell;** as bacterial cell **lacks splicing machinery;** not having any introns within its own genes; **Functional** foreign protein can thus be expressed by bacterial host; since **cDNA contains only coding sequence;**

(c) Why is it important to assess the environmental effects of growing GM crops? [2]

Discussion of ecological implications e.g.

possibility of **transfer of introduced herbicide resistance gene from GM crops to wild relatives;** through **cross-pollination;** resulting in development of “**superweeds**”; which are difficult to eradicate; need to assess possible effects of transgene products, like toxins, on other organisms; e.g. endangered wild species;

(d) With reference to **Table 4.1,**

(i) state which genetically modified crop seemed to be less harmful to the environment. [1]

GM maize;;

(ii) give an explanation for your answer in (d)(i). [2]

Weeds / surface invertebrates increase for GM maize compared to GM oilseed rape;; indicating that the presence of the proteins to confer resistance to herbicides have not hindered the growth and proliferation of other non-threatening organisms like surface invertebrates;;

(iii) discuss which is a more commercially viable crop for the farmer to grow. [2]

GM oilseed rape;;

due to decreased weed density or decreased number of invertebrates pests (relate these to increased crop yield);;

OR

Lower crop losses due to reduced competition for nutrients with weeds; And due to less damage by reduced numbers of invertebrate pests;

OR

Lower costs of herbicides to kill off weeds; and costs of pesticides to kill invertebrate pests;

(c) Suggest how the transfer could have taken place. [1]

Transgene present in pollen grains; Through pollen transfer; from GM maize / GM oilseed rape to other plants resulting in pollination; and fertilization;

[Total: 12]

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Answers:

Figure 2 is an electron micrograph which shows the axon of a motor neurone in transverse section.

(a)

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**ions are taken into the neurone via active transport.
ATP is required for the action of Ca²⁺ pumps which actively transport Ca²⁺ ions out**

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Label the structures X and Y. [2]

X: axoplasm Y: myelin sheath

(b)

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**of the neurone.
ATP is used for the transport / movement of synaptic vesicles towards the
membrane of the synaptic knob for the release of neurotransmitter.
ATP used for the absorption / endocytosis of choline and acetate back into the**

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Explain the significance of Y in electrical transmission of nervous impulses.

[3]

Y acts as an electrical insulator, preventing the loss of current from the axon

during electrical transmission.

In a myelinated neurone, action potentials are regenerated only at the nodes of

Ranvier / propagated by saltatory conduction, thus increasing the speed of electrical transmission.

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Often present at the synaptic knob of many motor neurons

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are a large number of mitochondria.

(c)(i)

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State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor

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neurone. [2]

Any two:

ATP is required for the Na⁺/K⁺ pump to work. Na⁺ ions are pumped out while K⁺

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(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

[2]

During intense activity, more action potentials are generated;
Increased activities of Na^+ - K^+ pumps needing more ATP

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(d) Suggest the effect on an action potential when the external concentration of sodium ions are lowered.

[2]

Less sodium ions enter the neurone;
Hence lower action potential

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(d) (i)

High Na^+ concentration outside the cell and a high K^+ concentration inside the cell;
Quote any cell's values –

Location	Ions	Ions concentration in			
		Frog muscle	Human r.b.c	Squid axon	<i>Chaetomorpha</i> (marine alga)
Outside the cell	Na^+	114	150	400	500
	K^+	14	14	14	14
Inside the cell	Na^+	14	14	42	42
	K^+	135	150	300	400

[Each point → 1 mark]

(ii)

Cell membrane;

Centered

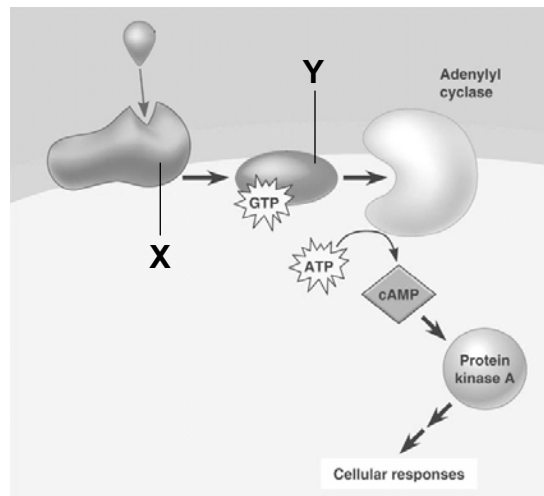
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Cell Signaling SAJC Prelim 2007 **QUESTION 8**

Certain microbes cause disease by disrupting G-protein signaling pathways. The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human feces colonizes the small intestine and produces a toxin that modifies a G protein.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salts into the intestines, leading to profuse diarrhea and death from loss of water and salts.



(a) Name molecules **X** and **Y**.

X **Y**[1]

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways?

.....[1]

(c) What is the significance of a phosphorylation cascade in cell signaling?

.....

.....

.....

.....

.....[2]

(d) State the function of adenylyl cyclase.

.....[1]

(e) Suggest how the toxin leads to an increase in cAMP concentration.

.....

.....

.....

.....

.....[3]

[Total: 8]

(a) Name molecules X and Y. [1]

[½ mark per pt]

X = G-protein linked receptor

Y = G-protein

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways? [1]

second messengers

(c) What is the significance of a phosphorylation cascade in cell signaling? [2]

signal amplification

specificity of cell signaling

ref. cross-talk (interaction between different signaling pathways)

(d) State the function of adenylyl cyclase. [1]

convert ATP to cAMP

(e) Suggest how the toxin leads to an increase in cAMP concentration.

[3]

modified G protein unable to hydrolyze GTP to GDP
remains stuck in its active form
adenylyl cyclase is continuously stimulated to make cAMP

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Photosynthesis and Respiration
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NADP/ NAD as

- Electron/hydrogen carrier or coenzymes;
 - Carry hydrogen atoms from one compound to another;
- [max1, ½ m each]

NADP in photosynthesis

- Final electron acceptor in non-cyclic photophosphorylation;
 - Drives photolysis of water;
 - $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
 - NADPH passes to Calvin cycle/ light independent reactions;
 - in the stroma;
 - NADPH reduces glycerate biphosphate to triose phosphate and gets reoxidized back to NADP;
- [max2, ½ m each]

NAD in respiration

Aerobic respiration:

Glycolysis: Accept H^+ from triose phosphate, which is oxidized to pyruvate;

Forms 2 NADH for each glucose molecule;

Link reaction: Accept H^+ from pyruvate, which is converted to acetyl CoA;

Forms 2 NADH for each glucose molecule;

Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;

Forms 6 NADH for each glucose molecule;

Link rxn and Krebs cycle occurs in the mitochondrial matrix

Carry hydrogen/ electrons to ETC;

Oxidative phosphorylation/ transfer of electrons down the ETC to

release ATP;
 1 NADH yields 3 ATP;
 NAD reoxidised;

Anaerobic respiration:

NAD regenerated when electrons from NADH transferred to pyruvate/
 ethanal;

NAD reused in glycolysis to generate 2 ATP;

[max5, ½ m each]

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(c) Explain the effect of carbon dioxide and light as limiting factors on the rate of [6]

photosynthesis.

CO₂ is a raw material for the calvin cycle

Its increased concentration increased the rate of photosynthesis

Its level is low (0.04%) in the atmosphere and therefore plays a major role in limiting photosynthesis

Light intensity is also important in the light dependent stage to excite electrons

Chl a absorbs max light at 680nm while chl b absorbs max light at 480nm

Compensation point is the light intensity when rate of photosynthesis and respiration is the same and it is reached at low light intensity

Is therefore rarely limiting during daylight hours

High light intensity also opens the stomata

Photosynthesis

Photosystems / antennae complex / chlorophyll in thylakoid membranes;

ETC, stalked particles (ATP synthase) in thylakoid membranes;

Thylakoid membranes are ion-impermeable / ref. to H⁺ gradient;

Ref. to thylakoid membrane separating compartments (compartmentalization);

Ref. to double membrane / envelope;

Ref. to enzymes and Calvin cycle in the stroma;

[max 3]

Respiration

Folding of cristae to increase surface area so that there are more stalked particles;

More ETC for oxidative phosphorylation;

ETC and stalked particles in inner membrane / cristae;

Inner membrane ion impermeable / ref. to proton gradient;

Ref. to intermembrane space acting as reservoir of H⁺;

Ref. to outer membrane as a barrier;

Ref. to enzymes and Krebs cycle in matrix

[max 3]

Anatomical homology

anatomical resemblances representing variations on a structural theme present in their common ancestor

ref. examples:

forelimbs of all mammals, including humans, cats, whales and bats

show the same arrangement of bones from the shoulder to the tips of the digits/ vestigial legs in snakes

even though these appendages can have very different functions (lifting, walking, swimming and flying)

Embryology homology

comparison of early stages of animal development reveals additional anatomical homologies not visible in adult organisms

ref. examples:

embryos of different types of vertebrates have a tail posterior to the anus/ pharyngeal (throat) pouches.

these embryonic structures develop into homologous structures with very different functions, such as gill slits in fishes and parts of ears and throat in humans.

Molecular homology

all forms of life use the same genetic machinery of DNA and RNA, and the genetic code is essentially universal.

DNA differences are accumulated as descendants evolve / closely related species share a greater portion of their DNA

ref. examples:

organisms as dissimilar as humans and bacteria share many genes inherited from a distant common ancestor

/ comparison of amino acid sequence of human haemoglobin with other vertebrates reveals the same pattern of evolutionary relationships as nonmolecular methods

In relation to Darwin's theory

ref. similarities between individuals of the same species or among different species, that arose from common ancestry

NOTES:

Darwin's theory = many modern species of organisms are descendants of ancestral species that were different from the species. As the descendants of that ancestral organism spilled into various habitats over millions of years, they accumulated diverse modifications, or adaptations, that fit them to specific ways of life.

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(b)

RJC Promo 2006

Distinguish between DNA replication and transcription.

Or RJC Prelim 2007

Describe the structure and function of the portions of eukaryotic DNA that do not encode protein or RNA. [6]

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Describe the neutral theory of molecular evolution.

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Explain how a point mutation in the gene coding for a growth factor receptor can lead to cancer.

OR

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Explain the role of telomeres in maintaining the integrity of a eukaryotic organism.

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Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells.

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FEATURE	DNA REPLICATION	TRANSCRIPTION	
Enzyme involved	DNA polymerase	RNA polymerase	[1]
Enzyme binds at?	DNA polymerase binds to parental DNA molecule at the origin of replication	RNA polymerase binds to the promoter of the gene to be transcribed	[1]
Raw material	Deoxyribonucleotides	Ribonucleotides	[1]
	Adenine, thymine, cytosine, guanine	Adenine, uracil, cytosine, guanine	[1]
Template	Both strands of DNA molecule act as template	Only 1 strand acts as template	[1]
	The whole DNA molecule is being replicated	Only a portion (gene) of the sense strand is being transcribed	[1]
Presence of proof-reading	DNA polymerase proofreads the newly synthesized daughter strand, ensuring correct complementary base-pairing.	Not present	[1]
Products	2 daughter DNA molecules	1 mRNA molecule	[1]
	Products remain in nucleus	Products leave the nucleus	[1]
Purpose	In cell division	Protein synthesis	[1]

[Any 5 – 1 mark each]

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Explain how changes at the genome level can lead to cancer.		
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Through genetic drift, these new neutral alleles become more common in the population and become a characteristic feature of the population;		
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10 (b) Explain how a point mutation in the gene coding for a growth factor receptor can lead to cancer. [6]		

Mutation in the proto-oncogene that codes the growth factor receptor, results in it becoming an oncogene/ becoming oncogenic
A point mutation results in a change in triplet code/ which changes the amino acid sequence in the growth factor receptor.
The mutated growth factor receptor will have different conformation and function differently.
The mutated receptor might be constitutively activated regardless of the presence of its ligand
OR the mutated receptor might be constitutively active because the mutated receptor does not dissociate from its ligand readily.
This results in persistent intracellular signaling for cell growth and proliferation leading to uncontrolled cell division and formation of tumors

the increase in cell division increased chances of multiple mutations in other critical genes like tumor suppressor genes, DNA repair genes, cell cycle control genes
This results in cells becoming cancerous

Cancerous cells metastasize /spread to other parts of the body to form secondary tumors.

(Cancerous cells can deprive surrounding cells/organs of nutrients and oxygen and cause functional failure of invaded tissues.)

OR

Explain the role of telomeres in maintaining the integrity of a eukaryotic organism. [6]

Are non-coding DNA sequences found at the ends of chromosomes.

Prevents fusion between exposed ends of chromosomes - chromosomal mutation.

Which can disrupt regulatory control of genes on the adjoined chromosomes

Prevents the activation of DNA repair enzymes which will recognize any exposed chromosomal ends as a defect and degrade the chromosomes.

serves as buffer DNA to protect the organism's genes from being eroded after successive rounds of DNA replication

so that critical proteins will still be synthesized in the daughter cells despite the shortened chromosomes

Cell will also undergo apoptosis after a limited number of cell division/mitosis (~50)/when a critical length of the telomere is reached

This limits the extent of accumulated mutations and prevents the development of cancer.

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